ARTMS Target Position Feedback System

THE UNIVERSITY OF BRITISH COLUMBIA
School of Engineering

Group I32
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Introduction

ARTMS develops and manufactures radioisotope technologies for cyclotron facilities and hospitals, utilizing pneumatic transport systems for isotope-carrying capsules. However, these capsules can sporadically jam in tight, lead-shielded conduits. This makes them difficult to locate, causing production downtime and delaying patient treatments.

We developed a distributed sensor checkpoint system for ARTMS to provide real-time positional feedback of capsule travelling inside the transfer hose.

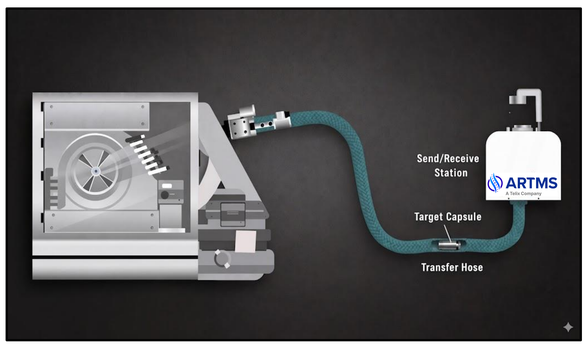

Fig. 1. Cyclotron Interface

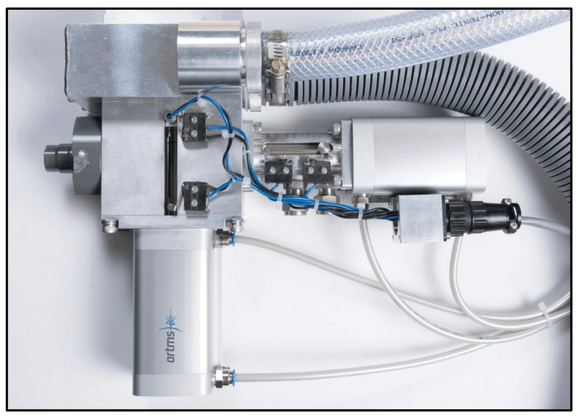

Fig. 2. ARTMS Isotope Manufacturing Technology

Methodology

A custom interface connected an air compressor to the hose to simulate capsule transfer. Time-of-flight (ToF), accelerometer, and inductive sensors were tested to evaluate detection performance and determine the most reliable checkpoint sensing method.

The sensor's data is read by the PLC through the sensor and interface PCBs. HMI is used to provide live capsule tracking.

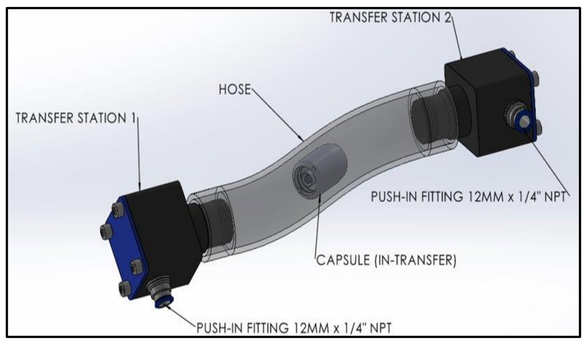

Fig. 3. Interface Assembly

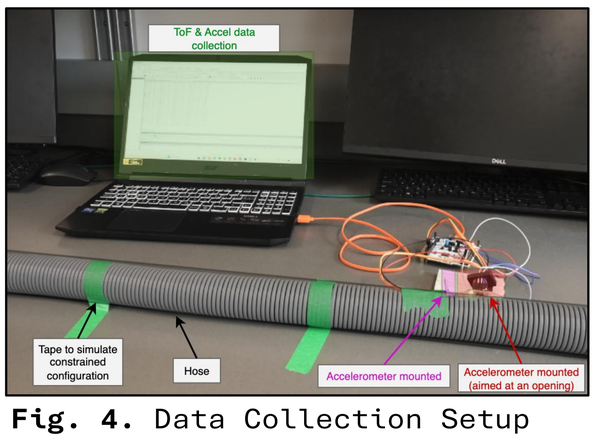

Fig. 4. Data Collection Setup

Results

The final design demonstrated a functional capsule detection and monitoring system, integrating pneumatic, electrical, and software components into one complete system.

System Components Overview:

- Pneumatic Test Bench
- Custom PCB
- Sensor Coupling
- HMI (Human Machine Interface)

Fig. 5. Test Bench Setup

Fig. 6. Test Bench Interface

Hose Test Bench Interface
The 3D-printed interface connected the hose to an air compressor, forming a sealed test bench that successfully propelled the capsule at ~80-110 PSI.

Sensor PCB
A custom PCB integrated redundant sensors into the checkpoint coupling, combining accelerometer detection with ToF verification; an inductive sensor was also evaluated.

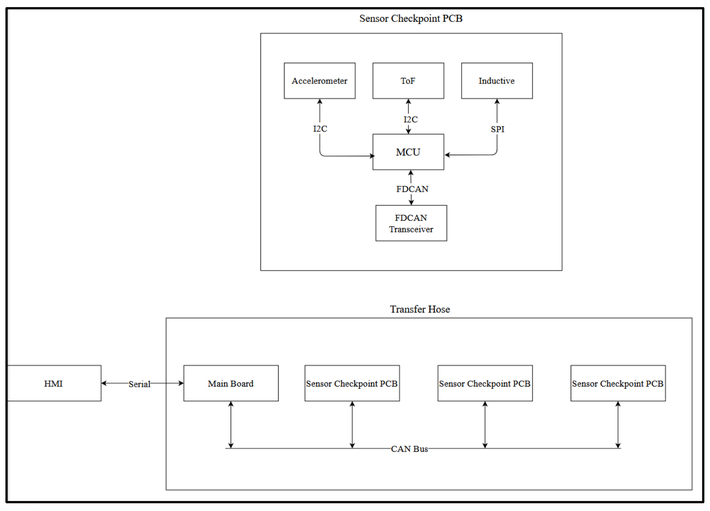

Fig. 7. System Architecture

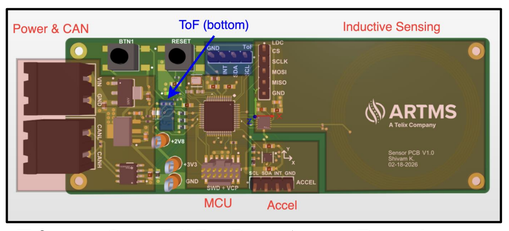

Fig. 8. PCB Design Render

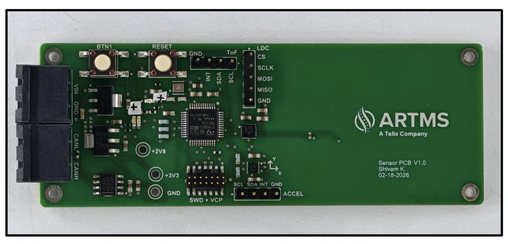

Fig. 9. Physical PCB

Sensor Data
Sensor data was collected under two test conditions: constrained and unconstrained hose configurations. These tests evaluated sensor performance under simulated operating conditions and realistic capsule transport scenarios. Both conditions resulted in distinct signal peaks as shown below. The results were consistent and robust.

Fig. 10. Constrained Hose Data

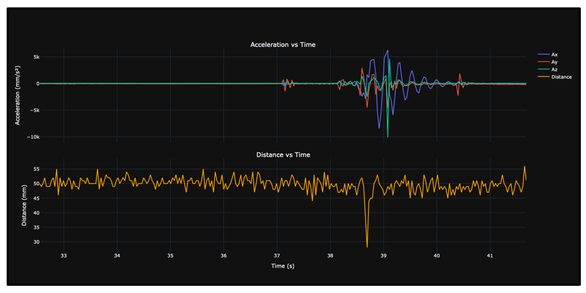

Fig. 11. Unconstrained Hose Data

Coupling
The coupling connects two hose sections and serves as a checkpoint housing for the sensor PCB. Two main designs were developed and evaluated.

The coupling performs the following:

- Connects hose sections
- Holds and mounts the PCB
- Acts as a checkpoint location

Final design choices:

- Transparent coupler for facilitating ToF sensing
- Polycarbonate (PC)-based part for machinability & low brittleness
- 10° ramp feature for friction fit with hose

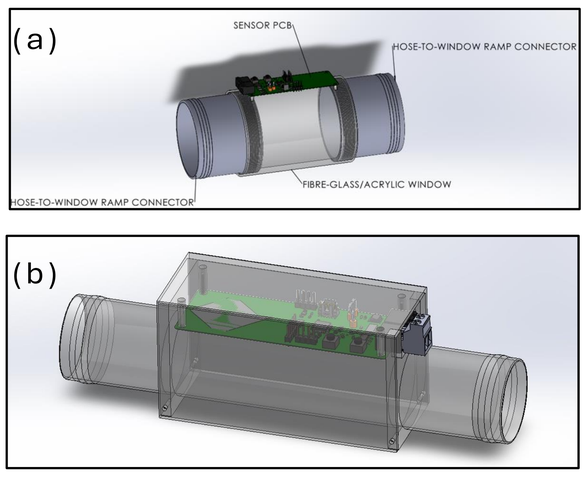

Fig. 12. Coupling Designs
(a) Transparent Coupling Design
(b) Enclosure Coupling Design

Human Machine Interface (HMI)
Two HMIs were developed (Siemens PLC and PC UI) communicating via CAN-to-USB. An interface hub PCB was designed, but a commercial CAN interface was used for demonstration.

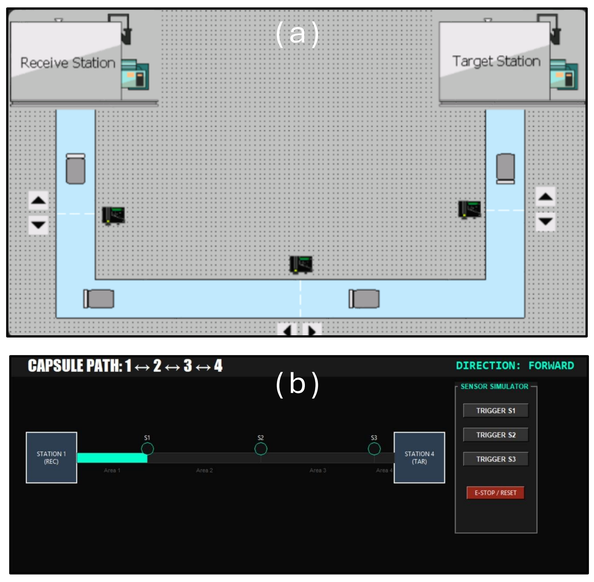

Fig. 13. System HMI Designs
(a) PLC Integration HMI
(b) Demonstration HMI

Conclusions

This project demonstrated a functional capsule detection checkpoint system and validated the use of embedded sensors integrated into a coupling-based system for capsule detection. The accelerometer and ToF sensors successfully detected capsule passage, demonstrating the feasibility of a checkpoint monitoring system for pneumatic capsule transport. This system can improve capsule tracking reliability and reduce retrieval time in cyclotron facilities. Significant experience was gained in mechanical design, PCB design, sensor testing, and system integration.

Future Improvements
Future work includes further accelerometer testing, development of a smaller PCB, and a sleeve-style checkpoint design using redundant accelerometer sensors to enable a less invasive installation.

References

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Ashhar, Z., Ahmad Fadzil, M. F., Othman, M. F., Yusof, N. A., Abdul Onny, M. A., Mat Ail, N., & Abd Rahman, S. F. (2023). Cyclotron production of Gallium-68 radiopharmaceuticals using the 68Zn(p,n)68Ga reaction and their regulatory aspects. Pharmaceutics, 15(1), 70. https://doi.org/10.3390/pharmaceutics15010070

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